a very long wire residing in free space leads to the result that the ***magnetic flux density*** **B** induced by a constant current I flowing in the z direction is given by

$$\mathbf{B} = \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r} \quad \text{(T)}, \tag{1.13}$$

where r is the radial distance from the current and $\hat{\boldsymbol{\phi}}$ is an azimuthal unit vector expressing the fact that the magnetic field direction is tangential to the circle surrounding the current. The magnetic field is measured in tesla (T), named in honor of Nikola Tesla (1856–1943). Tesla was an electrical engineer whose work on transformers made it possible to transport electricity over long wires without too much loss. The quantity μ_0 is called the ***magnetic permeability of free space*** [$\mu_0 = 4\pi \times 10^{-7}$ henry per meter (H/m)], and it is analogous to the electric permittivity ε_0. In fact, as we will see in Chapter 2, the product of ε_0 and μ_0 specifies c, which is the ***velocity of light in free space***:

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8 \quad \text{(m/s)}. \tag{1.14}$$

We noted in Section 1-3.2 that, when an electric charge q' is subjected to an electric field **E**, it experiences an electric force $\mathbf{F}_e = q'\mathbf{E}$. Similarly, if a charge q' resides in the presence of a magnetic flux density **B**, it experiences a ***magnetic force*** $\mathbf{F}_m$, but only if the charge is in motion and its velocity **u** is in a direction not parallel (or anti-parallel) to **B**. In fact, as we learn in more detail in Chapter 5, $\mathbf{F}_m$ points in a direction perpendicular to both **B** and **u**.

To extend Eq. (1.13) to a medium other than free space, μ_0 should be replaced with μ, which is the ***magnetic permeability*** of the material in which **B** is being observed. The majority of natural materials are ***nonmagnetic***, meaning that they exhibit a magnetic permeability $\mu = \mu_0$. For ferromagnetic materials, such as iron and nickel, μ can be much larger than μ_0. The magnetic permeability μ accounts for ***magnetization*** properties of a material. In analogy with Eq. (1.11), μ of a particular material can be defined as

$$\mu = \mu_r \mu_0 \quad \text{(H/m)}, \tag{1.15}$$

where μ_r is a dimensionless quantity called the ***relative magnetic permeability*** of the material. The values of μ_r for commonly used ferromagnetic materials are given in Appendix B.

▶ We stated earlier that **E** and **D** constitute one of two pairs of electromagnetic field quantities. The second pair is **B** and the ***magnetic field intensity*** **H**, which are related to each other through μ:

$$\mathbf{B} = \mu \mathbf{H}. \tag{1.16}$$

◀

1-3.4 Static and Dynamic Fields

In EM, the time variable t, or more precisely if and how electric and magnetic quantities vary with time, is of crucial importance. Before we elaborate further on the significance of this statement, it will prove useful to define the following time-related adjectives unambiguously:

- ***Static***—describes a quantity that does not change with time. The term ***dc*** (i.e., direct current) is often used as a synonym for static to describe not only currents, but other electromagnetic quantities as well.
- ***Dynamic***—refers to a quantity that does vary with time, but conveys no specific information about the character of the variation.
- ***Waveform***—refers to a plot of the magnitude profile of a quantity as a function of time.
- ***Periodic***—a quantity is periodic if its waveform repeats itself at a regular interval, namely its period T. Examples include the sinusoid and the square wave. By application of the Fourier series analysis technique, any periodic waveform can be expressed as the sum of an infinite series of sinusoids.
- ***Sinusoidal***—also called ***ac*** (i.e., alternating current), describes a quantity that varies sinusoidally (or cosinusoidally) with time.

In view of these terms, let us now examine the relationship between the electric field **E** and the magnetic flux density **B**. Because **E** is governed by the charge q and **B** is governed by $I = dq/dt$, one might expect that **E** and **B** must be somehow related to each other. As we learn later, they may or may not be interrelated, depending on whether I is static or dynamic.

Let us start by examining the dc case in which I remains constant with time. Consider a small section of a beam of charged particles, all moving at a constant velocity. The moving charges constitute a dc current. The electric field due to that section of the beam is determined by the total charge q contained in it. The magnetic field does not depend on q, but rather on the rate of charge (current) flowing through that section. Few charges moving very fast can constitute the same

磁场量与静态、时变场

（承上页）自由空间中，一条沿z方向流过恒定电流I的无限长直导线产生的磁通量密度为：

$$\mathbf{B} = \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r} \quad \text{(T)}, \tag{1.13}$$

r为到导线的径向距离；带帽号的φ为方位角单位矢量，表示磁场沿环绕电流的圆周切向。磁通量密度的单位是特斯拉（T），得名于电气工程师Nikola Tesla（1856—1943）。他在变压器等方面的工作推动了低损耗远距离输电。μ_0为真空磁导率，原书取$4\pi \times 10^{-7}$ H/m，它与介电常数ε_0相对应。两者的乘积决定真空光速：

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8 \quad \text{(m/s)}. \tag{1.14}$$

如第1-3.2节式（1.11）所述，电荷q′在外电场**E**中受到电场力$\mathbf{F}_e = q'\mathbf{E}$。处于磁场**B**中的电荷只有在运动且速度**u**不与**B**平行或反平行时，才受到非零磁场力$\mathbf{F}_m$。该力同时垂直于**B**和**u**，详见第5章。

在原书采用的均匀线性介质模型中，将式（1.13）的μ_0换为介质磁导率μ，可描述相应介质中的场。多数天然材料近似为非磁性材料，$\mu \approx \mu_0$；铁、镍等铁磁材料的μ可远大于μ_0。μ表征材料的磁化响应，常写为：

$$\mu = \mu_r \mu_0 \quad \text{(H/m)}, \tag{1.15}$$

μ_r为无量纲的相对磁导率，常用铁磁材料的值见附录B。第二对基本场量是**B**与磁场强度**H**，在此介质模型下满足：

$$\mathbf{B} = \mu \mathbf{H}. \tag{1.16}$$

1-3.4 静态场与时变场

电磁量是否以及如何随时间变化至关重要。为此约定以下术语：

静态：量不随时间变化。dc（直流）有时也用于描述不随时间变化的其他电磁量。
动态或时变：量随时间变化，但不限定变化的具体形式。
波形：量的数值随时间变化的曲线。
周期性：波形每隔周期T重复，例如正弦波和方波；满足通常傅里叶展开条件的周期波形可分解为正弦分量之和。
正弦变化：量随时间作正弦或余弦变化；本书以ac讨论此类激励。

电场与电荷q有关，磁场与电流I = dq/dt有关，两者是否耦合须区分静态和时变情形。先考虑恒定电流：取匀速带电粒子束的一小段，其中的总电荷量决定相应电场，而穿过截面的电荷流率（电流）决定相应磁场。少量电荷高速运动与大量电荷低速运动可以形成相同的电流。（接下页）

校注：交流并不必然是正弦波；此处限定为本书的正弦稳态讨论。光速3×10^8 m/s为近似值。2019年SI重新定义后，μ_0的数值不再精确等于$4\pi \times 10^{-7}$ H/m，但该近似足以用于本书的常规计算。BIPM：https://www.bipm.org/en/history-si/ampere 。

Table 1-3 The three branches of electromagnetics.

Branch	Condition	Field Quantities (Units)
Electrostatics	Stationary charges ($\partial q/\partial t = 0$)	Electric field intensity **E** (V/m) Electric flux density **D** (C/m^2) $\mathbf{D} = \varepsilon\mathbf{E}$
Magnetostatics	Steady currents ($\partial I/\partial t = 0$)	Magnetic flux density **B** (T) Magnetic field intensity **H** (A/m) $\mathbf{B} = \mu\mathbf{H}$
Dynamics (time-varying fields)	Time-varying currents ($\partial I/\partial t \neq 0$)	**E**, **D**, **B**, and **H** (**E**, **D**) coupled to (**B**, **H**)

current as many charges moving slowly. In these two cases, the induced magnetic field is the same because the current I is the same, but the induced electric field is quite different because the numbers of charges are not the same.

Electrostatics and ***magnetostatics*** refer to the study of EM under the specific, respective conditions of stationary charges and dc currents. They represent two *independent* branches, so characterized because the induced electric and magnetic fields do not couple to each other. ***Dynamics***, the third and more general branch of electromagnetics, involves ***time-varying fields*** induced by time-varying sources, that is, currents and associated charge densities. If the current associated with the beam of moving charged particles varies with time, then the amount of charge present in a given section of the beam also varies with time, and vice versa. As we will see in Chapter 6, the electric and magnetic fields become coupled to each other in that case.

► A time-varying electric field generates a time-varying magnetic field, and vice versa. ◄

Table 1-3 provides a summary of the three branches of electromagnetics.

The electric and magnetic properties of materials are characterized by the parameters ε and μ, respectively. A third fundamental parameter also needed is the ***conductivity*** of a material σ, which is measured in siemens per meter (S/m). The conductivity characterizes the ease with which charges (electrons) can move freely in a material. If $\sigma = 0$, the charges do not move more than atomic distances, and the material is said to be a ***perfect dielectric***. Conversely, if $\sigma = \infty$, the charges can move very freely throughout the material, which is then called a ***perfect conductor***.

Table 1-4 Constitutive parameters of materials.

Parameter	Units	Free-Space Value
Electrical permittivity ε	F/m	$\varepsilon_0 = 8.854 \times 10^{-12} \approx \frac{1}{36\pi} \times 10^{-9}$
Magnetic permeability μ	H/m	$\mu_0 = 4\pi \times 10^{-7}$
Conductivity σ	S/m	0

► The parameters ε, μ, and σ are often referred to as the ***constitutive parameters*** of a material (**Table 1-4**). A medium is said to be ***homogeneous*** if its constitutive parameters are constant throughout the medium. ◄

Concept Question 1-1: What are the four fundamental forces of nature, and what are their relative strengths?

Concept Question 1-2: What is Coulomb's law? State its properties.

Concept Question 1-3: What are the two important properties of electric charge?

(continued on p. 22)

电磁学分支与介质本构参数

（承上页）这两种粒子束的电流I相同，因而产生的磁场相同；但其电荷数量不同，电场就不同。静电学研究静止电荷，静磁学研究恒定电流。在这些条件下，相应的电场与磁场不通过时变项相互耦合，因而可作为两个独立分支研究。

更一般的电动力学研究时变源（电流及有关电荷密度）所产生的时变场。原书以带电粒子束为例讨论电荷量与电流的时间变化，并指出此时须考虑电场与磁场的耦合，详见第6章。

表1-3 电磁学的三个分支（按原书概括）

分支	源的条件	场量及关系
静电学	静止电荷；∂q/∂t = 0	电场强度 E（V/m） 电通量密度 D（C/m^2） D = εE
静磁学	恒定电流；∂I/∂t = 0	磁通量密度 B（T） 磁场强度 H（A/m） B = μH
电动力学（时变场）	时变电流；∂I/∂t ≠ 0	E、D、B、H 电场与磁场相互耦合

适用条件校注：上表是原书的入门分类，不是一般场问题的充分判据。由连续性方程，某体积内电荷的变化率取决于穿过其边界的净电流，而非某截面电流是否随时间变化。时变场的耦合应以麦克斯韦方程中的时间导数项理解，不能把“时变电场必然产生时变磁场，反之亦然”作为无条件命题。

介质的电学、磁学性质分别由ε、μ表征，还需要电导率σ，单位为西门子每米（S/m）。电导率描述电荷在介质中形成传导电流的能力。σ = 0对应理想电介质：电荷不能产生宏观传导，但仍可出现原子尺度的极化位移。σ → ∞是理想导体模型。

表1-4 介质的本构参数

参数	单位	真空值（原书近似）
介电常数 ε	F/m	$\varepsilon_0 \approx 8.854 \times 10^{-12} \approx 10^{-9}/(36\pi)$
磁导率 μ	H/m	$\mu_0 \approx 4\pi \times 10^{-7}$
电导率 σ	S/m	0

ε、μ和σ称为介质的本构参数。若本构参数在介质各处取值相同，则称该介质为均匀介质。

概念题

1-1：自然界有哪四种基本相互作用？原书给出的相对强度分别是多少？
1-2：什么是库仑定律？说明其性质。
1-3：电荷的两项重要性质是什么？

本节概念题续于原书第22页。

Technology Brief 1: LED Lighting

After lighting our homes, buildings, and streets for over 100 years, the incandescent light bulb created by Thomas Edison (1879) will soon become a relic of the past. Many countries have taken steps to phase it out and replace it with a much more energy-efficient alternative: the ***light-emitting diode*** (***LED***).

Light Sources

The three dominant sources of electric light are the incandescent, fluorescent, and LED light bulbs (**Fig. TF1-1**). We examine each briefly.

Incandescent Light Bulb

► ***Incandescence*** is the emission of light from a hot object due to its temperature. ◄

By passing electric current through a thin tungsten filament, which basically is a resistor, the filament's temperature rises to a very high level, causing the filament to glow and emit visible light. The intensity and shape of the emitted spectrum depends on the filament's temperature. A typical example is shown by the green curve in **Fig. TF1-2**. The tungsten spectrum is similar in shape to that of sunlight (yellow curve in **Fig. TF1-2**), particularly in the blue and green parts of the spectrum (400–550 nm). Despite the relatively strong (compared with sunlight) yellow light emitted by incandescent sources, the quasi-white light they produce has a quality that the human eye finds rather comfortable.

► The incandescent light bulb is significantly less expensive to manufacture than the fluorescent and LED light bulbs, but it is far inferior with regard to energy efficacy and operational lifetime (**Fig. TF1-7**). ◄

Of the energy supplied to an incandescent light bulb, only about 2% is converted into light, with the remainder wasted as heat! In fact, the incandescent light bulb is the weakest link in the overall conversion sequence from coal to light (**Fig. TF1-3**).



Figure TF1-1 (a) Incandescent light bulb; (b) fluorescent mercury vapor lamp; (c) white LED.

Fluorescent Light Bulb

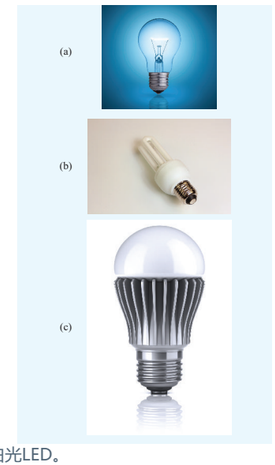
To ***fluoresce*** means to emit radiation in consequence to incident radiation of a shorter wavelength. By passing a stream of electrons between two electrodes at the ends of a tube (**Fig. TF1-1**(b)) containing mercury gas (or the noble gases neon, argon, and xenon) at very low pressure, the electrons collide with the mercury atoms, causing them to excite their own electrons to higher energy

技术简介1：LED照明

原书写道：自Thomas Edison于1879年研制实用白炽灯以来，白炽灯已照亮住宅、建筑物和街道一百多年，如今将逐步退出历史舞台。许多国家已采取措施，用更加节能的发光二极管（LED）替代白炽灯。

光源

本节比较三种主要电光源：白炽灯、荧光灯和LED灯，如图TF1-1。



图TF1-1 （a）白炽灯；（b）荧光汞蒸气灯；（c）白光LED。

白炽灯

白炽发光是物体因温度很高而发出可见光的现象。电流通过细钨丝（本质上为电阻元件）时，灯丝温度升高，从而发光。发射光谱的强度和形状取决于灯丝温度。图TF1-2的绿色曲线给出一个典型例子；与太阳光的黄色曲线相比，两者在蓝、绿光区域（400—550 nm）的谱形有相似之处。尽管白炽灯的黄光成分相对更强，其近似白光通常给人舒适的视觉感受。

白炽灯的制造成本低于荧光灯和LED灯，但光效与使用寿命均明显较低，见图TF1-7。原书估计，输入白炽灯的能量仅约2%转化为可见光，其余主要以热的形式散失。因此，在从煤的化学能到可见光的转换链中，白炽灯是效率最低的环节，见图TF1-3。

荧光灯

荧光是物质受到较短波长辐射激发后发射辐射的现象。荧光灯管中装有低压汞蒸气及有关工作气体。电子在灯管两端电极之间运动，并与汞原子碰撞，使原子的电子跃迁到较高能级。（接下页）

编校说明：本技术简介中的价格、寿命、光效及对2025年的预测均属于原书写作时的资料，不能视作当前产品参数。原书括号另列氖、氩、氙等气体；不同放电光源的工作气体与发光机制须分别理解。

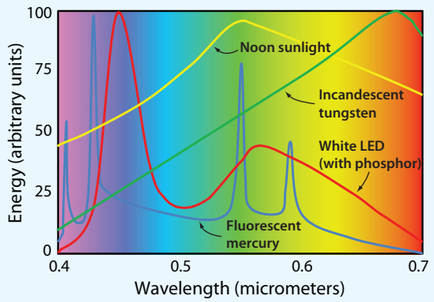


Figure TF1-2 Spectra of common sources of visible light.

levels. When the excited electrons return to the ground state, they emit photons at specific wavelengths, mostly in the ultraviolet part of the spectrum. Consequently, the spectrum of a mercury lamp is concentrated into narrow lines, as shown by the blue curve in **Fig. TF1-2**.

► To broaden the mercury spectrum into one that resembles that of white light, the inside surface of the fluorescent light tube is coated with phosphor particles [such as yttrium aluminum garnet (YAG) doped with cerium]. The particles absorb the UV energy and then reradiate it as a broad spectrum extending from blue to red; hence the name ***fluorescent***. ◄

Light-Emitting Diode

The LED contained inside the polymer jacket in **Fig. TF1-1**(c) is a p-n junction diode fabricated on a semiconductor chip. When a voltage is applied in a forward-biased direction across the diode (**Fig. TF1-4**), current flows through the junction and some of the streaming electrons are captured by positive charges (holes). Associated with each electron-hole recombining act is the release of energy in the form of a photon.

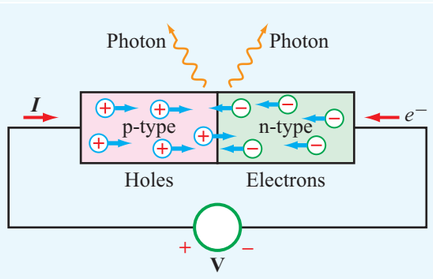


Figure TF1-4 Photons are emitted when electrons combine with holes.

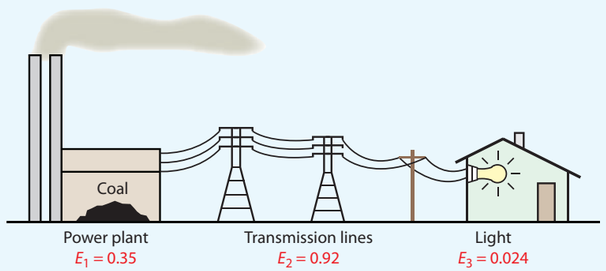
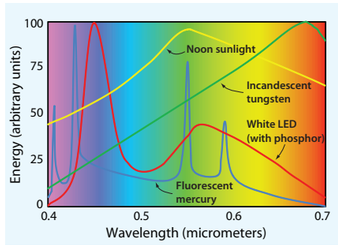


Figure TF1-3 Lighting efficiency. (Source: National Research Council, 2009.)

LED照明：荧光转换与电致发光

（承上页）受激电子回到基态时，会发射特定波长的光子，主要位于紫外波段。因此，汞灯光谱表现为若干窄谱线，见图TF1-2的蓝色曲线。



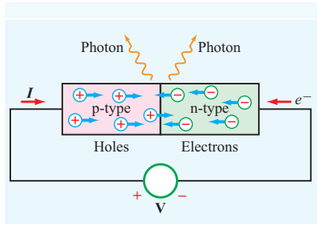
图TF1-2 常见可见光源的光谱。

图轴：横轴为波长（μm），纵轴为能量（任意单位）。曲线：正午太阳光、白炽钨丝灯、荧光汞灯，以及使用荧光粉的白光LED。任意单位用于比较谱形，不代表绝对辐射功率。

为把汞的谱线转换为更接近白光的宽光谱，灯管内表面涂覆荧光粉。荧光粉吸收紫外辐射，再发射较长波长、分布较宽的光，从蓝光延伸至红光，这就是“荧光灯”名称的由来。原书此处举出掺铈钇铝石榴石（YAG:Ce）作为荧光材料的例子。

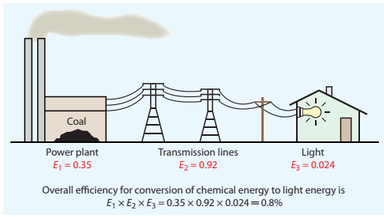
发光二极管

图TF1-1（c）中聚合物封装内的LED是在半导体芯片上形成的p-n结二极管。当结两端施加正向偏压时，电流通过结区；部分电子与空穴复合，并以光子的形式释放能量，见图TF1-4。



图TF1-4 电子与空穴的辐射复合产生光子。

图中Electrons：电子；Holes：空穴；Photon：光子；p-type、n-type：p型、n型。校注：并非每一次电子—空穴复合都发光，还存在非辐射复合；这里描述的是LED中的辐射复合机制。



图TF1-3 从煤的化学能到可见光的转换效率。

图TF1-3中，煤经发电厂、电力传输和灯泡转换为光；各环节效率分别为E_1 = 0.35、E_2 = 0.92、E_3 = 0.024。总效率为0.35 × 0.92 × 0.024 = 0.007728 ≈ 0.8%。资料来源：美国国家研究委员会，2009年。

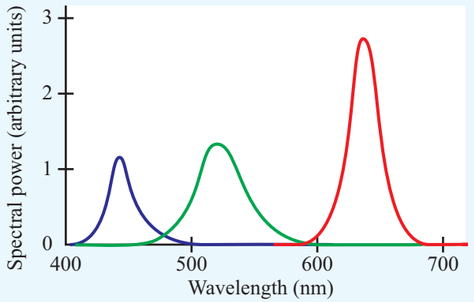


Figure TF1-5 The addition of spectra from three monochromatic LEDs.

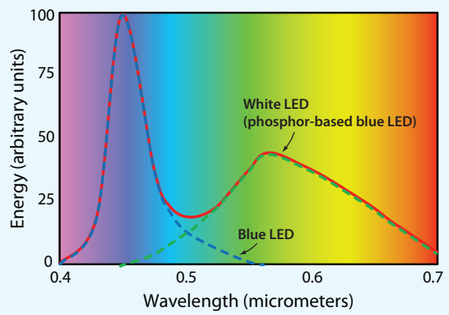


Figure TF1-6 Phosphor-based white LED emission spectrum.

► The wavelength of the emitted photon depends on the diode's semiconductor material. The materials most commonly used are aluminum gallium arsenide (AlGaAs) to generate red light, indium gallium nitride (InGaN) to generate blue light, and aluminum gallium phosphide (AlGaP) to generate green light. In each case, the emitted energy is confined to a narrow spectral band. ◄

Two basic techniques are available for generating white light with LEDs: (a) RGB and (b) blue/conversion. The RGB approach involves the use of three monochromatic LEDs whose primary colors (red, green, and blue) are mixed to generate an approximation of a white-light spectrum. An example is shown in **Fig. TF1-5**. The advantage of this approach is that the relative intensities of the three LEDs can be controlled independently, thereby making it possible to "tune" the shape of the overall spectrum to generate an esthetically pleasing color of "white." The major shortcoming of the RGB technique is cost: manufacturing three LEDs instead of just one.

With the blue LED/phosphor conversion technique, a blue LED is used with phosphor powder particles suspended in the epoxy resin that encapsulates it. The blue light emitted by the LED is absorbed by the phosphor particles and then reemitted as a broad spectrum (**Fig. TF1-6**). To generate high-intensity light, several LEDs are clustered into a single enclosure.

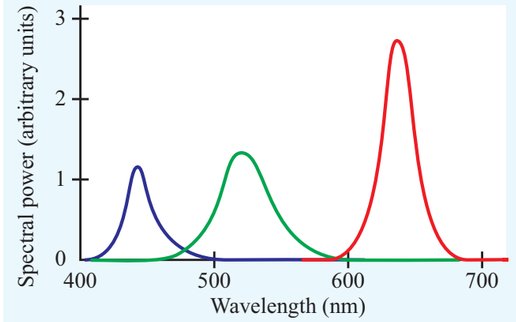
Comparison

► ***Luminous efficacy*** (***LE***) is a measure of how much light in lumens is produced by a light source for each watt of electricity consumed by it. ◄

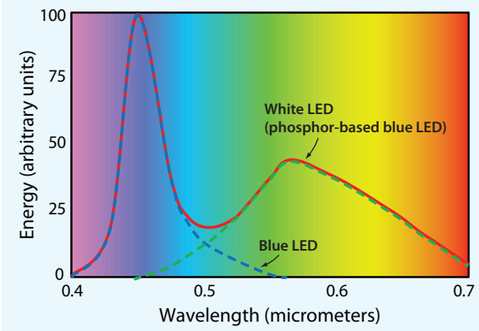
Of the three types of light bulbs we discussed, the incandescent light bulb is by far the most inefficient and its useful lifespan is the shortest (**Fig. TF1-7**). For a typical household scenario, the 10-year cost—including electricity and replacement cost—is several times smaller for the LED than for the alternatives.

白光LED与光效

LED发射光子的波长取决于半导体材料。原书列举：砷化铝镓（AlGaAs）用于产生红光，氮化铟镓（InGaN）用于蓝光，磷化铝镓（AlGaP）用于绿光。每一种单色LED的发射能量主要集中于较窄波段。



图TF1-5 三个单色LED的光谱相加。



图TF1-6 荧光粉转换型白光LED的光谱。

图TF1-5：横轴波长（nm），纵轴光谱功率（任意单位）。图TF1-6：横轴波长（μm），纵轴能量（任意单位）；曲线分别为蓝光LED与荧光粉转换型白光LED。化学式Al中的第二个字母是小写l（铝），不是大写I。

用LED产生白光有两种基本方法：（a）红、绿、蓝三色混光（RGB）；（b）蓝光激发荧光粉转换。RGB方法将三个单色LED的红、绿、蓝光混合，形成视觉上近似白光的输出，见图TF1-5。其优点是三色相对强度可以独立控制，从而调整总体光谱和白光色调。主要缺点是需制造、驱动三个LED，成本高于单LED方案。

蓝光LED / 荧光粉转换法把荧光粉颗粒分散在封装LED的环氧树脂中。部分蓝光被荧光粉吸收，再以较长波长的宽谱光重新发射，与剩余蓝光共同形成白光，见图TF1-6。要获得较高光输出，可以将多个LED集成于同一封装或灯具内。

比较

光效（luminous efficacy，LE）是光源输出的光通量与所消耗电功率之比，单位为流明每瓦（lm/W）。瓦特是功率单位，不能译成"每瓦电量"。

在原书比较的三类灯中，白炽灯光效最低、寿命最短，详见图TF1-7。按原书采用的家庭使用情景估算，将电费与更换费用合计，LED的10年使用成本仅为其他方案的一部分。

术语校注：光效（lm/W）与能量转换效率（无量纲）是不同物理量。RGB混光产生白色视觉感受，并不意味着其三峰光谱等同于太阳光的连续光谱。